SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 12

RAILWAY RESERVATION SYSTEM – NEGATIVE TEST SCENARIOS AND TEST CASES

Aim:

To Write a case study to identify and analyze the requirements of a Railway Reservation System and prepare negative test scenarios and test cases for the application.

1. Problem Description

A railway reservation system allows passengers to search trains, check seat availability, enter passenger details, select a class, pay reservation charges and receive a booking confirmation or ticket.

The Railway Reservation System is expected to provide reliable, secure and user-friendly services. The testing team verifies the stated requirements using valid and invalid inputs, records actual results, and reports any deviation from the expected behaviour.

2. Functional Requirements

1. Passengers must be able to search trains using source, destination and journey date.
2. The system must show train schedules and class-wise availability.
3. Passenger details must be validated before booking.
4. Seat allocation must not exceed available capacity.
5. Fare must be calculated according to route, class and applicable charges.
6. Booking is confirmed only after successful payment and seat allocation.
7. A confirmed booking must generate a unique PNR/ticket reference.
8. Users must be able to view or cancel bookings according to policy.

4. Negative Test Scenarios and Test Cases

Negative testing checks whether invalid, incomplete, unauthorized or boundary inputs are safely rejected with a clear message.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Search with missing source	Leave source station blank	Validation message is displayed.	Pass
NT02	Use same source and destination	Enter identical stations	Search/booking is rejected with a clear message.	Pass
NT03	Journey date in past	Select a past date	Past-date booking is not allowed.	Pass
NT04	Invalid passenger age	Enter negative or nonnumeric age	Passenger details are rejected.	Pass
NT05	Book with no availability	Select a class with zero seats	Booking is blocked or waitlist rules are shown.	Pass
NT06	Exceed passenger limit	Add more passengers than allowed per booking	Extra passengers are rejected.	Pass
NT07	Payment declined	Simulate payment failure	Ticket is not confirmed and payment status is shown.	Pass
NT08	Duplicate payment callback	Send repeated payment success callback	Only one booking/payment is recorded.	Pass

5. Detailed Execution of a Failed Test Case

Test Case ID: NT05

Precondition: The application is running in the test environment and the relevant test data is available.

Test Steps: 1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Select a class with zero seats. 4. Observe the application response.

Expected Result: Booking is blocked or waitlist rules are shown.

Actual Result: For demonstration, assume the application accepts the invalid input or does not display the required validation message.

Conclusion: The test is marked as failed for defect-report demonstration. This is an illustrative result, not a claim about a deployed application.

6. Sample Bug Report

Field	Description
Bug ID	BR-012
Linked Test Case	NT05
Severity	High (illustrative)
Priority	High (illustrative)
Bug Title	Railway Reservation System does not correctly validate the condition: Book with no availability.
Environment	Test environment; browser and operating system should be recorded during actual execution.
Preconditions	The application is running in the test environment and the relevant test data is available.
Steps to Reproduce	1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Select a class with zero seats. 4. Observe the application response.
Expected Result	Booking is blocked or waitlist rules are shown.
Actual Result	For demonstration, assume the application accepts the invalid input or does not display the required validation message.
Impact	Invalid data or an unauthorized operation may be accepted, resulting in incorrect processing, data integrity issues or a poor user experience.
Suggested Fix	Apply validation at both client and server sides, return a clear error message, and add regression tests for the boundary and invalid condition.
Status	Open – illustrative sample.

7. Bug Life Cycle

New → Assigned → In Progress → Fixed → Retesting → Closed. If the defect is still reproducible after the correction, the tester reopens it and returns it to the development team.

Conclusion

The test cases prepared for the Railway Reservation System provide coverage of its key functional requirements. Executing them helps identify functional gaps, improve error handling and increase confidence in the quality of the application.